



RAJALAKSHMI
ENGINEERING COLLEGE
An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

Continuous Assessment Test-I [CAT -I]

Year : I
Semester : II
Branch : B.E. / B.Tech.
Sub. Code : CS23231
Subject : Data Structures
QP Code : 011071

Date: 09.04.2025

[Regulations 2023]

Time: 120 Mints

Max. Marks: 75

(Common to CSE, ECE, EEE, IT, AIML, CSD, AI & DS, CS)

Answer All Questions

Part A: 10 x 2 = 20 Marks

- 1 Define self-referential structure and give example. [CO1] [A1]
- 2 How can you find the last element of a singly linked list? [CO1] [A1]
- 3 Consider a doubly linked list with nodes A, B, and C. If node B is deleted, what should be the new value of A's next pointer? [CO1] [A2]
- 4 What operation would be most efficient in a circular linked list compared to a singly linked list? And state why. [CO1] [A2]
- 5 A stack initially empty is subjected to a sequence of push and pop operations. The pushes push the integers from 1 to 10 in order, and the pop operations occur whenever an even integer is pushed. After all operations are completed, what integers are left in the stack? [CO2] [A2]
- 6 What value is returned by the pop() operation if the stack is empty? How is this condition typically handled? [CO2] [A2]
- 7 After performing three dequeue operations on a queue implemented with a linked list, what node becomes the new front if the queue initially contained the elements Q->R->S->T->U? [CO2] [A2]
- 8 Given a sequence of enqueue and dequeue operations on an initially empty queue, identify whether the resulting sequence is possible. Enqueue operations are denoted by E and dequeue operations by D. Sequence: E, E, D, D, E, D. Is this sequence of operations valid for a queue? Justify your answer. [CO2] [A2]
- 9 For a complete binary tree with n nodes, how many nodes have at least one child? [CO3] [A1]

- 10 Given a binary tree traversal of pre-order: A, B, D, E, C, F, G and in-order: D, B, E, A, F, C, G, reconstruct the tree. What node will be visited last in a post-order traversal? [CO3] [A2]

Part B: 5x11=55 Marks

- 11 a. Write a function in C to add a new node at the end of a singly linked list. Explain each step with neat diagram. [CO1] [B1]

[OR]

- b. Given two singly linked lists that represent numbers (in reverse order), write a function to add them and return the result as a new linked list. [CO1] [B2]

- 12 a. Write a function in C to insert a new node at the beginning, and end of a doubly linked list. [CO1] [B1]

[OR]

- b. Implement a function to insert a node at the end of a circular singly linked list. Explain how the circular nature is preserved. [CO1] [B2]

- 13 a. You are asked to design a simulation for a canteen plate rack system. In the rack, plates are stacked one over the other, and only the top plate can be removed at any time. This is a perfect real-life example of the LIFO (Last-In, First-Out) principle of stacks. [CO2] [B2]

Write a C program to simulate the following operations using a stack:

- push_plate(int plateNumber) → to add a plate to the rack
- pop_plate() → to remove the top plate
- display_stack() → to display the current arrangement of plates

Perform the following operations and show the stack content after each step:

- Add plates: 101, 102, 103
- Remove one plate
- Add plate: 104
- Display all plates

[OR]

- b. You are developing a calculator application that can evaluate expressions in postfix notation (also known as Reverse Polish Notation). Unlike infix expressions, postfix expressions do not require parentheses, and evaluation can be efficiently done using a stack. [CO2] [B2]

Evaluate the postfix expression:

6 5 2 3 + 8 * + 3 + *

- Show the stack content after each step.
- Clearly indicate the operands used and the intermediate result at each operation.

- 14 a. Discuss the conditions for overflow and underflow in an array-based queue. How do you detect and handle these conditions in a program? [CO2] [B2]

[OR]

- b. You are developing a basic ticket booking system for a small theatre where customers line up in a First-Come, First-Served manner. You decide to implement this queue using a linear array. The operations needed are: [CO2] [B2]

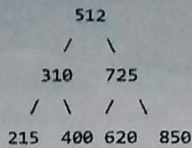
- enqueue() to add a new customer to the queue.
- dequeue() to serve the customer at the front of the queue.

Write the algorithm for enqueue() and dequeue() operations in a linear queue using arrays.

- 15 a. You are developing a feature for a new educational software that visually represents various data structures to students. One of the core features is to allow students to visualize how different tree traversals operate on a Binary tree. The BST is used to hold student IDs in a sorted manner for quick access and manipulation. [CO3] [B2]

Given Tree:

Imagine the software currently displays the following BST with student IDs:



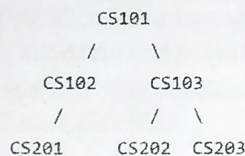
Implement functions to display the student IDs using each of the following tree traversals, in-order, pre-order and explain in comments within your code or in accompanying documentation how each traversal affects the order in which student IDs are displayed.

[OR]

- b. You have been hired to design a Course Prerequisite Management System for a university. The system must model course dependencies using a tree-based structure, where:

- Each node represents a course.
- The root is the foundational course (no prerequisites).
- Each child node is a course that requires the parent node as a prerequisite.

The following structure is provided:



Tasks:

1. Identify and define the following tree terminologies based on the above structure:
 - Root
 - Leaf nodes
 - Internal nodes
 - Degree of a node
 - Height of the tree
 - Parent and child
2. What is the degree of node CS103?

Determine whether the course tree is a full binary tree. Justify your answer using tree terminology.

Reg. No.

2 1 1 6 2 3 1 0 0 1 2 5 5



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Continuous Assessment Test- I [CAT - I]

Year : I
 Semester : II
 Branch : B.E./B.Tech.
 Sub. Code : CS23231
 Subject Name : Data Structures
 QP Code : 071002

[Regulations 2023]

[Common to CSE, IT, AI&ML, CSD, AI&DS and CSE (Cyber Security)]

Date: 23.04.2024

Time: 120 Minutes

Marks: 75

Answer ALL Questions

Part A [4 x 2 = 8 Marks]

- | | | |
|--|------|-----|
| 1.1. Define a self-referential structure. | [A1] | CO1 |
| 1.2. How is dynamic memory allocation used in linked lists? | [A2] | CO1 |
| 1.3. List the types of linked lists and one application for each. | [A1] | CO1 |
| 1.4. How does a circular linked list differ from a singly linked list? | [A2] | CO1 |

Part B [2 x 11 = 22 Marks]

- | | | |
|--|------|-----|
| 1.5. a. Explain with a diagram how dynamic memory allocation works and discuss its advantages. | [B1] | CO1 |
|--|------|-----|

[OR]

- | | | |
|---|------|-----|
| b. Illustrate the process of inserting and deleting nodes in a doubly linked list with examples. | [B2] | CO1 |
| 1.6. a. Describe the implementation of a circular linked list and explain its advantages over a singly linked list with examples. | [B1] | CO1 |

[OR]

- | | | |
|--|------|-----|
| b. Explain the concept of self-referential structures and demonstrate with code how they are used in linked list implementation. | [B2] | CO1 |
|--|------|-----|

Answer ALL Questions

Part A [4 x 2 = 8 Marks]

- | | | |
|---|------|-----|
| 2.1. What are the primary operations of a stack? | [A1] | CO2 |
| 2.2. How are stacks implemented using arrays? | [A2] | CO2 |
| 2.3. Define a queue and its main operations. | [A1] | CO2 |
| 2.4. How does linked list implementation of a queue differ from array implementation? | [A2] | CO2 |

Part B [2 x 11 = 22 Marks]

- 2.5. a. Discuss the method to evaluate the postfix expression "9 3 / 2 * 7 +". [B1] CO2
Provide a step-by-step breakdown of the operations, including the stack changes and the intermediate results after each operation.

[OR]

- b. Explain with a diagram the array implementation of a stack. [B2] CO2

- 2.6. a. Discuss the linked list implementation of queues. [B2] CO2

[OR]

- b. Explain circular queue using array implementation. [B2] CO2

Answer ALL Questions

Part A [2 x 2 = 4 Marks]

- 3.1. What are the methods to traverse a binary tree? [A1] CO3

- 3.2. Define a binary search tree and its unique property. [A2] CO3

Part B [1 x 11 = 11 Marks]

- 3.3. a. Discuss tree traversals with examples. [B1] CO3

[OR]

- b. Discuss the concept of binary search trees and illustrate with a diagram how insertion and deletion operations are performed. [B2] CO3

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Reg. No.

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Continuous Assessment Test-I [CAT -I]

Year : I
Semester : II
Branch : B.E /B.TECH
Sub. Code : CS23231
Subject : Data Structures
QP Code : 012071

(Regulations 2023)

Date: 09.04.2025

Time : 120 Min

Max. Marks : 75

(Common to ECE, IT, CSD, AI&DS)

Answer All Questions

Part A: 10 x 2 = 20 Marks

- 1 Write the C structure definition for a node in a singly linked list. [CO1] [A1]
- 2 State how the free() function works in C. What happens if free() is called more than once on the same memory address? [CO1] [A2]
- 3 How would you insert a new node between two existing nodes in a doubly linked list? Provide the steps [CO1] [A2]
- 4 How can you convert a singly linked list into a circular linked list? Provide the steps involved. [CO1] [A2]
- 5 What minimum number of stack operations (push or pop) are required to reverse the order of elements in a stack of size n? [CO2] [A1]
- 6 Show the state of the stack after processing the following operations: push('a'), push('b'), push('c'), pop(), push('d'). [CO2] [A1]
- 7 What is the result in the queue after performing the following sequence of operations: enqueue(1), enqueue(2), dequeue(), enqueue(3), enqueue(4), dequeue()? [CO2] [A1]
- 8 State how a queue differs from a stack in terms of data access and structure. What are the typical use cases for each data structure? [CO2] [A2]
- 9 Define the term "degree" in the context of a tree. How does the degree of a node affect its structure in a tree? [CO3] [A1]
- 10 Define the term "sibling" in tree terminology. How do siblings in a tree differ from siblings in a binary tree? [CO3] [A1]

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Part B: 5x11=55 Marks

- 11 a. Write a function in C to add a new node at the front of a singly linked list. Explain each step with neat diagram and discuss how the head pointer is managed. [CO1] [B1]

[OR]

- b. Write a C function to dynamically allocate a Float 1D array using malloc. After allocation, prompt the user to input elements into this array and then print the array elements. [CO1] [B1]

- 12 a. Describe the process of removing a node from the end of a doubly linked list. Include pseudocode and explain with examples and diagrams. [CO1] [B2]

[OR]

- b. Write pseudocode for subtracting two polynomials using singly linked lists. Include considerations for handling negative results and zero coefficient terms. [CO1] [B2]

- 13 a. You are enhancing a calculator app with support for evaluating postfix expressions (Reverse Polish Notation). Since postfix expressions eliminate the need for parentheses and operator precedence rules, they are ideal for stack-based computation. [CO2] [B2]

As part of your development task, you are required to write a function that evaluates postfix expressions using a stack and simulate its behavior step-by-step.

Problem Statement:

1. Write a C function to evaluate a postfix expression using a stack.
 - o Operands are positive integers.
 - o Supported operators: +, -, *, /.
2. Evaluate the following postfix expression using your function: 7 4 5 * + 6 2 / -

[OR]

- b. You are building a simple code editor for students learning programming. One of the essential features is to validate whether all brackets in a code snippet are balanced. The editor should support checking the three common types of brackets used in programming: [CO2] [B2]

- Parentheses ()

- Curly Braces {}
- Square Brackets []

To perform this validation efficiently, you decide to implement the logic using a **stack data structure**, since it follows the **Last-In, First-Out (LIFO)** principle — perfectly suited for matching nested opening and closing brackets.

Design an algorithm using a stack to check whether an expression contains balanced brackets.

- Push every opening bracket (, {, [onto the stack.
- When a closing bracket is encountered, pop the top of the stack and check for a match.
- If the stack is empty or mismatched at any point, the expression is not balanced.

Write a C function to implement this algorithm.

Using your function, check whether the following expressions have balanced brackets:

- {[()}]

- 14 a. Implement a queue using arrays that can handle integer data. [CO2] [B1]
Include functions for enqueue, dequeue, and displaying all elements of the queue.

[OR]

- b. Design a linked list-based stack. Write a function push() to add an element to the stack and pop() to remove the top element. Describe how the linked list changes with each operation. [CO2] [B1]

- 15 a. Define and contrast the terms 'complete binary tree', 'full binary tree', and 'perfect binary tree'. Provide examples where each type might be used in software applications. [CO3] [B1]

[OR]

- b. You're assigned to develop a **directory management feature** for an operating system simulation tool used in a computer science course. The file system is represented as a **binary tree**, where: [CO3] [B2]

- Each **node** represents a folder.
- The **left child** represents a subfolder with a name lexicographically smaller than the parent.
- The **right child** represents a subfolder with a name lexicographically larger than the parent.

The current folder structure is as follows:

```
Documents
 /      \
Assignments projects
 /      \      \
Class1 class2 Thesis
```

- i. Write a function to print the folder names in lexicographical order.
- ii. What will be the output of the in-order traversal on this tree?
- iii. How is this traversal useful when users want folders listed alphabetically?